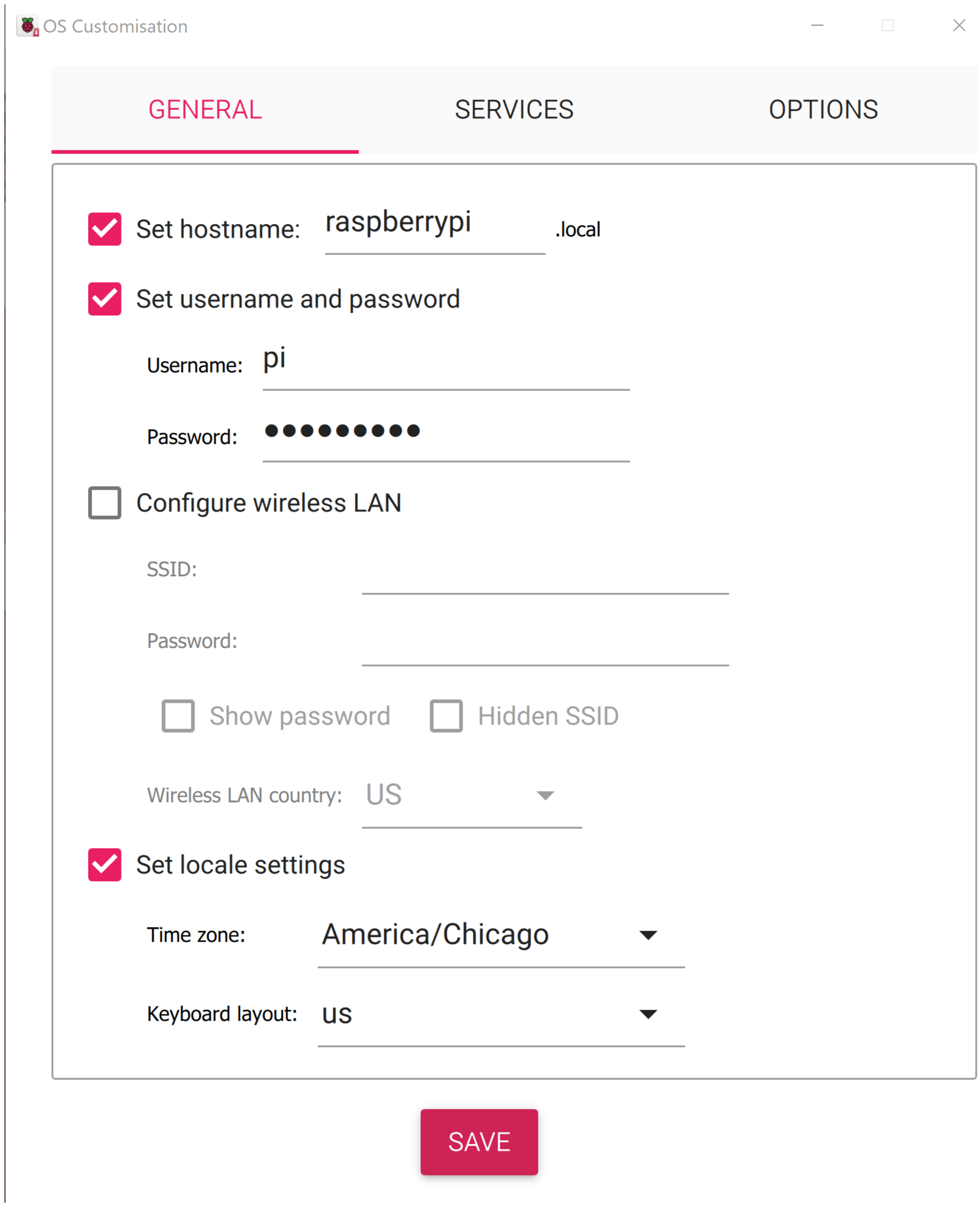
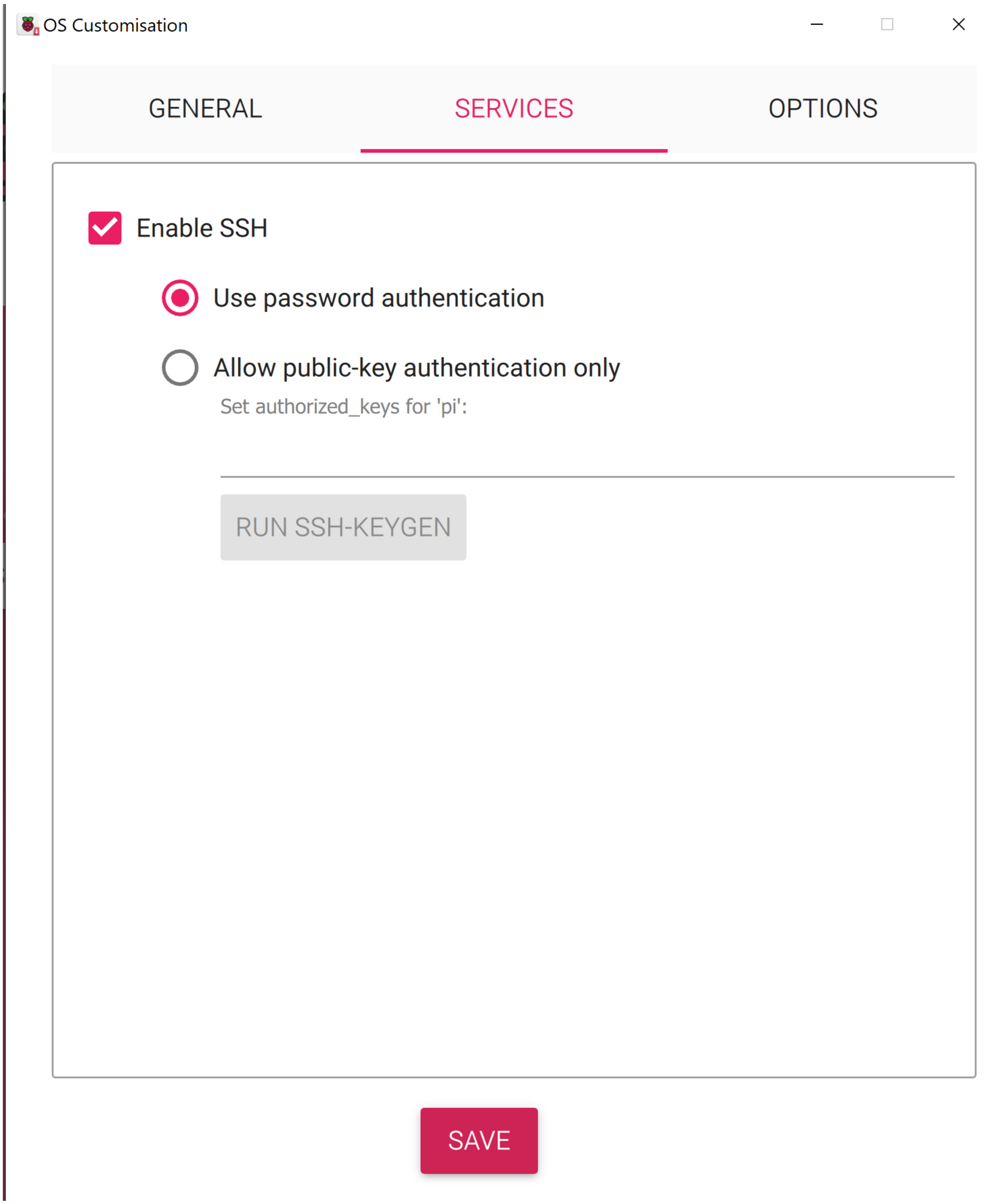
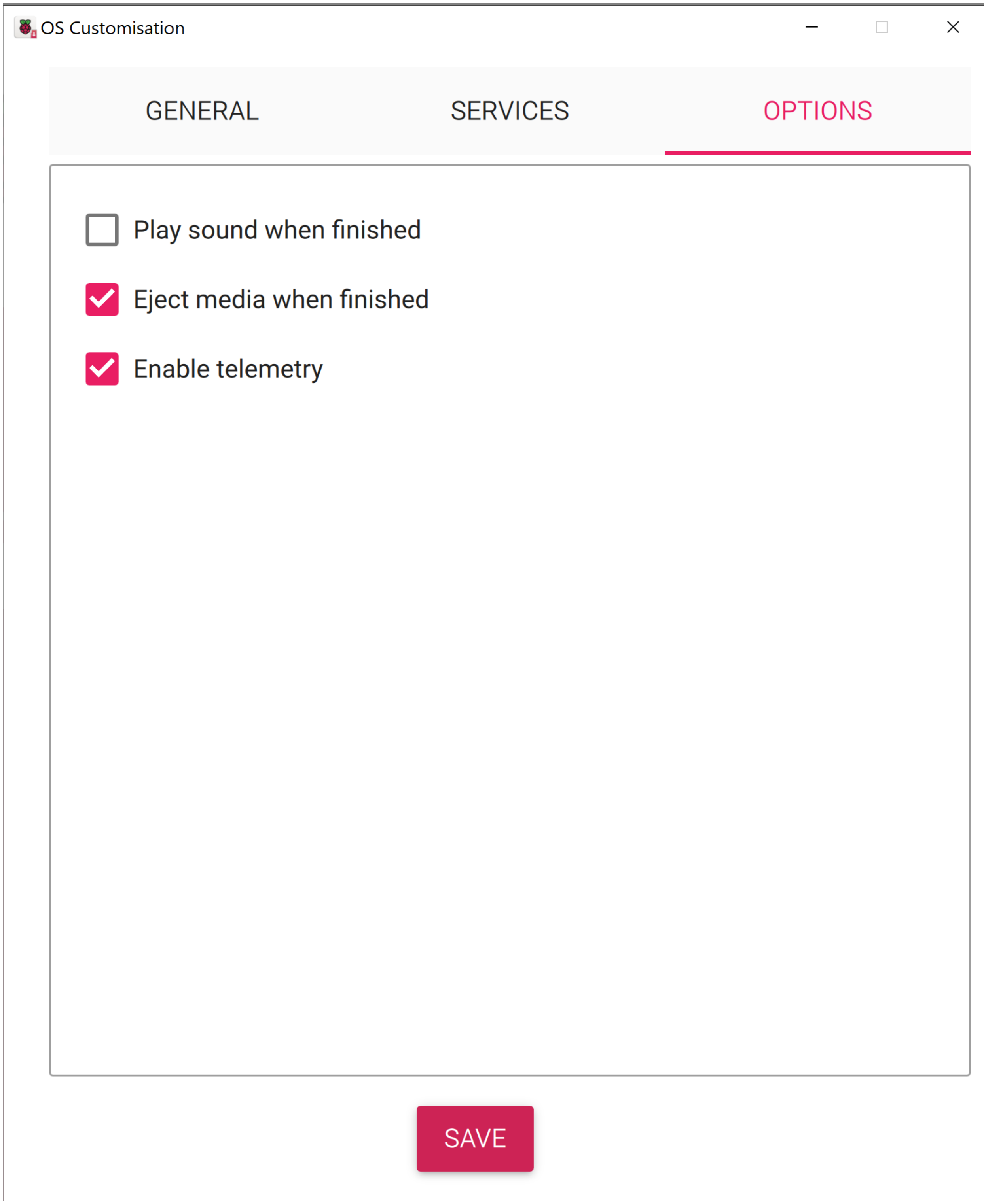
Raspbian OS 64-bit on PiRacer Setup Guide

Downloading and installing the Raspbian OS image:

1. Download the image using [this link.]
2. Download and install Raspberry Pi Imager.
3. Run Raspberry Pi Imager and do the following:
   a. Click ‘Choose device’ and select ‘Raspberry Pi 4’.
   b. Click ‘Choose OS’ and select ‘Use custom’, then navigate to and select the image you downloaded earlier.
   c. Click ‘Choose storage’ and select the SD card.
   d. Click ‘Next’.
   e. When prompted to apply OS customization settings, click ‘Edit Settings’ and set the following settings (the password is raspberry):


OS Customisation
GENERAL
SERVICES
OPTIONS
Set hostname:
raspberrypi
.local
Set username and password
Username:
pi
Password:
Configure wireless LAN
SSID:
Password:
Show password
Hidden SSID
Wireless LAN country:
US
Set locale settings
Time zone:
America/Chicago
Keyboard layout:
us
SAVE


OS Customisation
GENERAL
SERVICES
OPTIONS
Enable SSH
Use password authentication
Allow public-key authentication only
Set authorized_keys for 'pi':
RUN SSH-KEYGEN
SAVE



4. Wait for the image to flash to the SD card, then eject it from your computer.
    a. If you plan to use SSH, open the bootfs drive and create an empty file called ‘ssh’ with no extension.

5. Put the SD card into the Raspberry Pi and turn it on. It may or may not boot up with the battery pack alone yet, so plug it in to a power source if needed.

Setting up Raspbian and installing dependencies:

1. Connect the RPi to a monitor and keyboard/mouse, or SSH into it from your computer.
2. If you are prompted to run through a setup, complete that and also make sure the RPi is connected to the internet.
3. Open a terminal and run the following commands:
```
sudo apt-get update --allow-releaseinfo-change
sudo apt-get upgrade
```
4. Launch the raspi-config utility:
```
sudo raspi-config
```
   a. Enable 'Interfacing Options – I2C'
   b. Select Advanced Options – Expand Filesystem
   c. Choose <Finish> and hit enter. Reboot after changing these settings.

Setup the Virtual Environment:

1. Upgrade gast before creating the virtual environment:
```
sudo pip install -U gast --break-system-packages
```
2. To create a virtual environment run the following:
```
python3 -m venv env --system-site-packages
echo "source ~/env/bin/activate" >> ~/.bashrc
source ~/.bashrc
```
3. Install required libraries:
```
sudo apt install libcap-dev libhdf5-dev libhdf5-serial-dev
```

Install Donkeycar Python Code:

1. Run:
```
pip install donkeycar[pi]
```
2. Then run:
```
sudo apt install python3-opencv
```

Install OLED Display Service:

1. Before installing the service for the OLED display, we need to downgrade the version of setup tools currently installed on the RPi.
2. First, deactivate the virtual environment by running:
```
deactivate
```
3. Then, run the following to downgrade setuptools to the correct version:
```
sudo pip install setuptools==46.1.1 --break-system-packages
```
4. Reactivate the virtual environment:
```
source ~/.bashrc
```
5. Run the following commands to install the OLED display service:
```
cd ~
git clone https://github.com/waveshare/pi-display
cd pi-display
sudo ./install.sh
```
6. You should now see the OLED display turn on!

Create DonkeyCar:

1. Create donkeycar example by running the following:
```
donkey createcar --path ~/mycar
```

Ensure Everything is Working:

1. To check if the camera is working:
   a. Remove the lens cap from the camera lens.
   b. Run the following:

```
sudo rpicam-jpeg --output test.jpg --timeout 2000 --width 640 --height 480
```

   c. If it doesn't work, run these commands and try again:
```
sudo apt update
sudo apt upgrade
```
   d. If the rpicam-jpeg command runs with no errors, there will be a test.jpg file with a picture taken from the camera.

2. To check if Tensorflow is installed correctly:
```
python -c "import tensorflow; print(tensorflow.__version__)"
```

    a. If the version is printed, it has been installed correctly!
3. To check if OpenCV was installed correctly:

```
python -c "import cv2"
```

    a. If nothing happens and there are no errors, it has been installed correctly!

If you followed this guide exactly, you shouldn't run into any issues. If you do, reach out to me on Discord!